Supplementary Material for *Lateral Phishing as a Violation of the Organizational Communication Graph: Cascade Dynamics and Leak-Aware Evaluation*

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This supplement contains secondary tables and figures referenced from the main text. All values are produced by the reproducibility package (`code/results/*.csv`); none are new experiments.

S1. Detector hyperparameters

Model / component	Hyperparameters (fixed, without tuning)
GNN (static / TGN-lite / TGAT-lite)	DIM=24; TGAT: t_{dim}=8; 100 epochs; Adam lr=0.01, wd=$5{\cdot}10^{-4}$; dropout 0.2
GNN classifier head	MLP, 48 hidden units
Tabular detectors (LightGBM)	n_estimators=200–250, seed=42, num_leaves=31, lr=0.1
volumetric / path-based (LightGBM)	n_estimators=200, seed=42; context window W=2 buckets

Table S1: Detector hyperparameters: fixed *uniformly* for all models of a given family, framework defaults or lightly chosen, without grid search.

S2. Cascade detectors and the cascade-model sweep

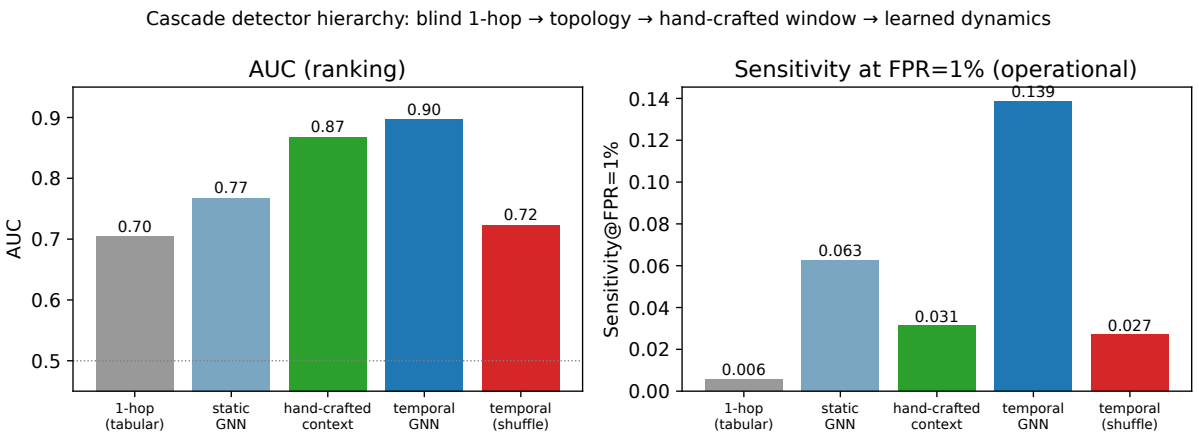


Figure S1: Detector hierarchy on AUC (left) and sensitivity@FPR1% (right); values as in the cascade-detector table of the main text. The temporal GNN dominates; time shuffling (red) levels it with the static GNN.

p_{inf}	0.05	0.1	0.2	0.3	0.5	0.7	0.9
1-hop	0.60	0.56	0.59	0.60	0.68	0.73	0.79
manual	0.89	0.87	0.85	0.84	0.85	0.87	0.89
temporal	**0.96**	**0.96**	**0.91**	**0.88**	**0.88**	**0.90**	**0.93**

Table S2: Cascade model sweep (AUC, depth 4, 8 seeds): the hierarchy holds across the whole range of the infection probability, and the receiving-side signal does not vanish at a realistically low p_{inf}.

S3. Inductive validation under increasing scale

	AUC		sensitivity@FPR1%	
Organizations	temp.	manual	temp.	manual
19	0.83	0.86	**0.123**	0.028
30	0.80	0.86	**0.087**	0.054
200	0.84	0.88	**0.107**	0.077

Table S3: Inductive validation on unseen organizations (19 organizations: 5 seeds; 30 and 200 organizations: 3 seeds): the temporal detector's operational advantage over the hand-crafted context holds on companies never seen in training, but shrinks with scale ($4.4\times \rightarrow 1.4\times$).

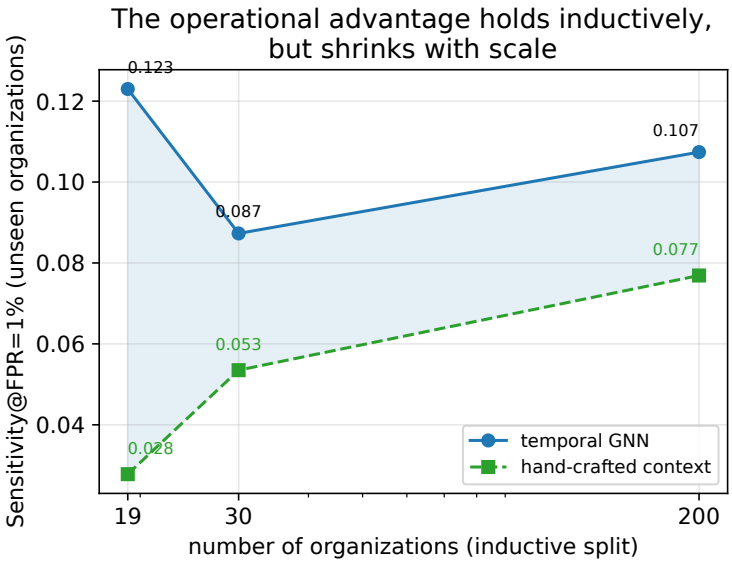


Figure S2: Inductive validation: sensitivity@FPR1% on unseen organizations against the number of companies (values as in Tab. S3). The advantage holds inductively but decreases ($4.4\times \rightarrow 1.4\times$).

S4. Replication on real topologies

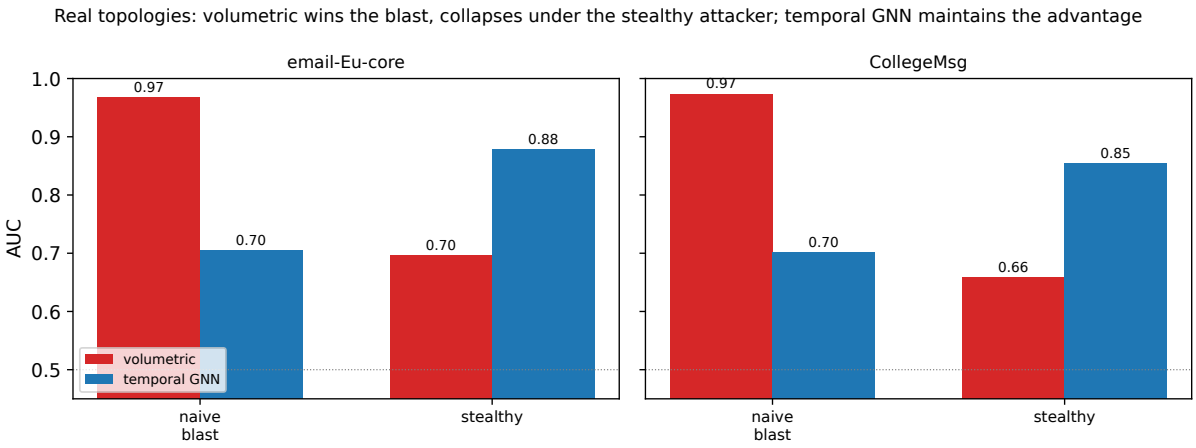


Figure S3: Two real topologies (`email-Eu-core`, `CollegeMsg`). The volume detector (red) wins the naive blast, but drops under the stealthy attack; the temporal GNN (blue) stays higher. Values as in block (b) of the real-graph table of the main text.

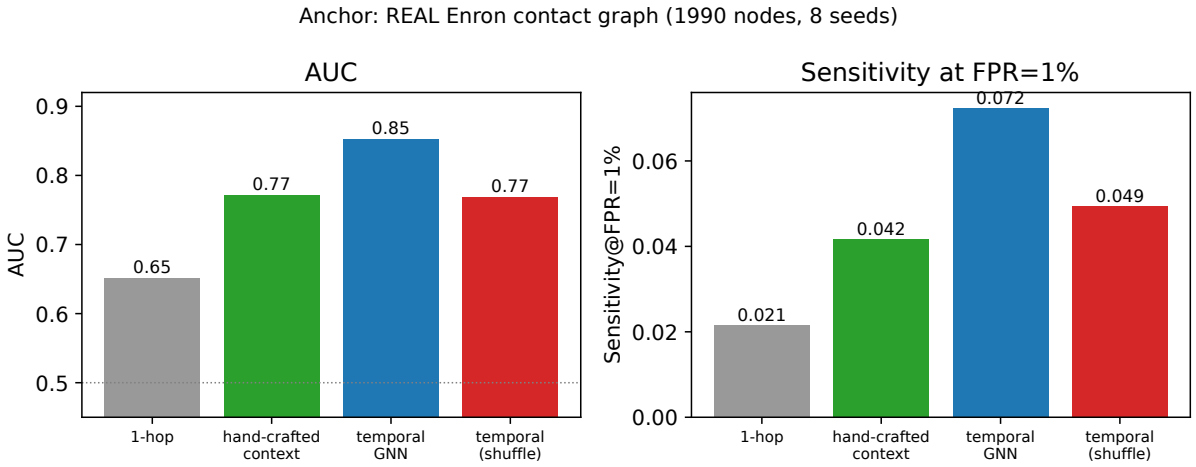


Figure S4: The cascade invariant on Enron (values as in block (a) of the real-graph table of the main text): the same hierarchy as on the case-study graph (1-hop $<$ hand-crafted context $<$ temporal GNN), with the time shuffle leveling the temporal detector to the hand-crafted result.

S5. Rigor: label noise and the sensitivity–FPR curve

Label noise	vol.	path	manual	temp.
0%	0.79	0.72	0.66	**0.82**
5%	0.78	0.71	0.65	**0.80**
10%	0.78	0.71	0.64	**0.76**
20%	**0.75**	0.68	0.62	0.70

Table S4: Audit of noisy training labels (AUC, stealthy attack; 20 seeds): the learned temporal model degrades fastest, so its advantage shrinks under realistic label noise.

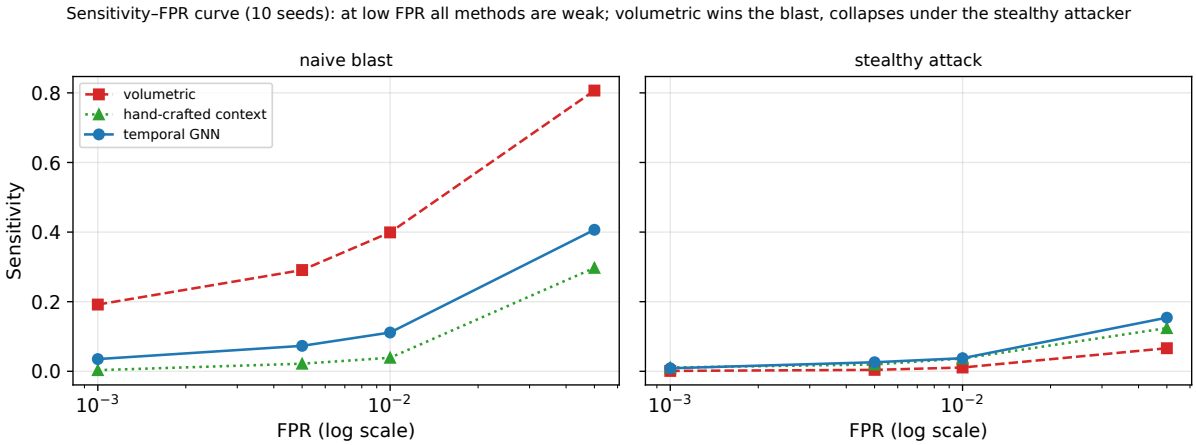


Figure S5: Sensitivity–FPR curve (10 seeds). At low FPR all methods are weak; the volume detector dominates the blast and breaks down under the stealthy attack, where the receiving-side signal degrades less.

S6. Feature specification of every detector

All tabular detectors use the same learner (LightGBM, 200–250 trees as in Tab. S1, seed 42); they differ only in their features. Time is divided into buckets b; the context window is $W{=}2$ buckets. $\mathcal{N}(v)$ is the set of contacts of v in the communication graph; recv(b) and sent(b) are the sets of accounts that received, respectively sent, mail in bucket b; rhythm(s, b) is the temporal-consistency bit *tc*. The two experiment harnesses of the package (`experiments/` for the cascade, stealth, scale, and real-graph tables; `cascadebench/` for the baseline, label-noise, and headline-contrast tables) implement identical feature definitions.

- **1-hop (tabular).** For event $s \to v$ in bucket b: $\mathbf{1}[s \in \mathcal{N}(v)]$, rhythm$(s, b)$, $|\mathcal{N}(v)|$, $|\mathcal{N}(s)|$.
- **Hand-crafted cascade context.** The 1-hop features plus three receiving-side statistics: *burst* $= |(\mathcal{N}(v) \setminus \{s\}) \cap \bigcup_{b'=b-W}^{b} \text{recv}(b')|/|\mathcal{N}(v)|$ (how many of the victim's other contacts were hit within the window); *freshness* $= \mathbf{1}[s \in \bigcup_{b'=b-W}^{b-1} \text{recv}(b')]$ (the sender was itself hit just before); *propagation* $= |(\mathcal{N}(v) \setminus \{s\}) \cap \bigcup_{b'=b-W}^{b} \text{sent}(b')|/|\mathcal{N}(v)|$ (how many of the victim's other contacts are themselves sending).
- **Volumetric baseline (inspired by COMPA).** Three per-event deviations of the sender from its own profile, computed on the observed stream: recipient novelty $1/(1 + \#\{s \to v\})$, time novelty $1/(1 + \#\{\text{events of } s \text{ in bucket } b\})$, and sender activity $\#\{\text{events of } s\}$. COMPA itself detects compromised accounts on social networks by comparing each new message with a behavioral profile of the account (time of day, source, language, links, topics); we transfer only its per-sender-deviation idea to email headers, restricted to volume and timing.
- **Path-based baseline (inspired by Hopper).** *Precursor* $= \mathbf{1}[s \in \bigcup_{b'=b-W}^{b-1} \text{recv}(b')]$ (the sender was reached just before, i.e. is part of a chain); *momentum* = number of distinct targets of s in buckets $[b, b + W]$; *rare target* $= 1/(1 +$ popularity of v as a recipient$)$; *unexplained* $= 1 -$ rhythm(s, b); and the conjunction precursor $\times$ unexplained. Hopper itself detects lateral movement in enterprise networks from authentication logs, by reconstructing login paths between machines and flagging paths with no benign explanation; it does not process email. We transfer its causal-path idea (a precursor that explains the hop, and a rare target) to the email communication graph.
- **Static GNN.** Node features $X = \log(1 + [\,|\mathcal{N}(u)|,\ \text{in-degree}(u),\ \text{out-degree}(u)\,])$ (three dimensions, from the event stream), projected to DIM=24 and propagated through two layers with the normalized adjacency (Eq. (1) of the main text). The head g is an MLP with 48 hidden units and dropout 0.2 over $[Z_s, Z_v, Z_s \odot Z_v, x_{sv}]$, where x_{sv} is the 1-hop feature vector of the event.
- **Temporal GNN (TGN-lite).** Same node features and x_{sv}; a memory of dimension 24 per node, updated bucket by bucket with a GRU cell from the mean of the messages $\sigma(W_m m_u^{(b-1)})$ of the senders that reached the node in bucket b (Eq. (3)); an event of bucket b is scored from the memories of bucket $b-1$ (Eq. (2)) by the same head as the static GNN. TGAT-lite replaces the mean pooling by attention with an 8-dimensional time encoding. Training: Adam (lr=0.01, weight decay $5 \cdot 10^{-4}$), 100 epochs, binary cross-entropy.